

NetworkCausalTree: An R package for heterogeneous treatment and spillover effects under network interference

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Summary

Interference occurs when treatment assigned to one unit influences the outcomes of other units (Cox, 1958). In the realm of policy interventions, interference can manifest through various interactions, encompassing social, physical, or virtual connections. The conventional Rubin Causal Model, employed in causal inference studies (Rubin, 1986), excludes interference. However, when interference is present, it must be taken into account to prevent a biased estimation of the treatment effect (Forastiere et al., 2016) and to be able to estimate spillover effects, that is, the effects on a unit's outcome of the treatment received by other units. Consequently, recent research has devised new methodologies to tackle interference on networks (Aronow & Samii, 2017; Athey et al., 2018; Forastiere et al., 2016; Leung, 2020). Concurrently, alongside the interference research domain, scholars have crafted machine learning algorithms to appraise treatment effect heterogeneity with respect to individual characteristics (Athey & Imbens, 2016). These algorithms identify heterogeneous sub-populations by iteratively segregating groups whose estimate conditional average treatment effect differs the most from others, and enabling researchers to understand which sub-populations are particularly responsive to the treatment.

To integrate the aforementioned two topics in the field of causal inference, Falco J. Bargagli-Stoffi et al. (2025) introduces a novel machine learning algorithm, named Network Causal Tree (NCT), that explores the heterogeneity of treatment and spillover effects concerning individual, neighborhood, and network characteristics within randomized settings. Evaluating the heterogeneity of treatment and spillover effect is crucial for identifying both those sub-populations who are responsive to the treatment they received and those who are highly susceptible to the treatment received by other units. Heterogeneity exploration is performed via an interpretable decision tree algorithm. Decision trees provide high levels of interpretability by offering transparent insights into the relationships in the data that are learned by the model (Murdoch et al., 2019). By recursively partitioning the data via decision rules, decision trees transform complex data patterns into an intelligible series of *if-then* decision rules. NCT is designed to operate within clustered neighborhood interference (CNI), where units are partitioned into distinct clusters, spillover mechanisms exclusively take place within clusters based on the links of a cluster-specific network and units are assumed to receive an indirect exposure to the intervention if they have at least one treated peer within their immediate neighbors. The estimation of conditional effects is carried out using an extended version of the Horvitz-Thompson estimator (Aronow & Samii, 2017), tailored to accommodate clustered network interference. NetworkCausalTree is an R Package providing a flexible implementation

of the Network Causal Tree algorithm.

This paper introduces the NetworkCausalTree R package, the first open-source implementation of a decision-tree method for estimating heterogeneous treatment and spillover effects under clustered network interference. All analyses and figures in this paper can be reproduced using the scripts and example data provided in the package's GitHub repository.

Statement of need

Existing R packages and algorithms are designed either to detect and estimate Heterogeneous Treatment Effects (HTEs) in the absence of interference or to estimate average causal effects in the presence of interference.

On the one hand, there are a plethora of methods to discover and estimate HTEs employing decision trees in the absence of interference. Some of them have also been implemented in R. For instance: i) the aVirtualTwins package, which implements the Virtual Twins algorithm introduced by (Foster et al., 2011); ii) the causalTree package, which implements the Causal Honest Tree algorithm introduced by Athey and Imbens (Athey & Imbens, 2016); iii) the policytree which implements the Policy Tree methodology as proposed by (Athey & Wager, 2021); iv) the CRE package (Cadei et al., 2023), which implements the Causal Rule Ensemble algorithm introduced by (Falco J. Bargagli-Stoffi et al., 2020).

On the other hand, there are three main packages to estimate average causal effects in the presence of interference: i) the interference package implements the inverse probability weighting estimator proposed by Tchetgen & VanderWeele (2012) to evaluate the causal effect of an intervention in the presence of partial interference in observational studies; ii) the interference package estimates causal effects under network interference in randomized experiments (Aronow & Samii, 2017); iii) the interference latenetwork package (Hoshino & Yanagi, 2024) estimates the causal effect of an intervention under noncompliance and unknown network interference.

Existing methods for causal inference either handle treatment effect heterogeneity *without* interference or interference *without* heterogeneity. The table below summarizes this gap:

Package	Handles Interference	Handles Heterogeneity	Network Structure
aVirtualTwins (Foster et al., 2011)	No	Yes	No
causalTree (Athey & Imbens, 2016)	No	Yes	No
policytree (Athey & Wager, 2021)	No	Yes	No
CRE (Falco J. Bargagli-Stoffi et al., 2020)	No	Yes	No
interference (Aronow & Samii, 2017)	Yes	No	Yes
interference (Tchetgen & VanderWeele, 2012)	Yes	No	Yes
NetworkCausalTree (ours)	Yes	Yes	Yes

NetworkCausalTree fills this hole by providing the first interpretable, decision-tree-based

72 approach for discovering heterogeneity in both treatment and spillover effects under network
73 interference.

74 To account for the current lack of existing methods for a data-driven discovery of
75 the heterogeneity of causal effects in the presence of network interference, we propose
76 NetworkCausalTree, an R package providing a flexible implementation of the Network
77 Causal Tree algorithm. NetworkCausalTree provides i) an interpretable representation of the
78 heterogeneity of treatment and spillover effects in the presence of CNI, within randomized
79 settings; ii) the possibility for the researcher to simultaneously model the heterogeneity of both
80 treatment and spillover effects. To the best of our knowledge, there aren't existing packages
81 to explore, in a data-driven manner, the heterogeneity of treatment and spillover effects in the
82 presence of network interference.

83 Algorithm

84 The NCT algorithm is designed to detect and estimate heterogeneous treatment and spillover
85 effects in randomized settings under CNI. Under CNI, spillover is confined to units within
86 the same cluster, and an individual's outcome is influenced by the treatment status of
87 directly connected units in the cluster-specific network. Potential outcomes $Y_{ik}(w, g)$ —where
88 $k \in \mathcal{K} = [1, \dots, K]$ is the cluster indicator, and $i = 1, \dots, n_k$ is the unit indicator in
89 each cluster k —are indexed with respect to the individual treatment $W_{ik} = w$ and to the
90 neighborhood treatment $G_{ik} = g$, which is defined as a binary variable that equals 1 if the unit
91 ik has at least one treated neighbor, 0 otherwise. For each observation ik in the sample $\mathcal{V}$ we
92 observe the cluster indicator M_{ik} and a vector of individual or network characteristics $\mathbf{X}_{ik}$.
93 The adjacency matrix describing interactions between units within a given cluster k is denoted
94 with $\mathbf{A}_k$. Estimands of interest $\tau_{(w, g; w', g')}$ encompass two treatment and two spillover effects,
95 comparing average potential outcomes under varying individual or neighborhood treatment
96 statuses, while keeping the other constant. For the subset of units with covariate vectors
97 $\mathbf{x} \in \mathcal{X}$ that are mapped in the leaf $l(\mathbf{x})$, we define the leaf-specific conditional average causal
98 effect (CACE) comparing the treatment status (w, g) and the one (w', g') as follows:

$$\begin{aligned} \tau_{(w, g; w', g')}(l(\mathbf{x})) &= \frac{1}{N(l(\mathbf{x}))} \sum_{k=1}^K \sum_{i=1}^{n_k} Y_{ik}(w, g) I(\mathbf{X}_{ik} \in l(\mathbf{x})) \\ &\quad - \frac{1}{N(l(\mathbf{x}))} \sum_{k=1}^K \sum_{i=1}^{n_k} Y_{ik}(w', g') I(\mathbf{X}_{ik} \in l(\mathbf{x})) \end{aligned}$$

99 where $N(l(\mathbf{x}))$ is the number of units within the leaf $l(\mathbf{x})$.

100 These effects will be estimated by a Horvitz-Thompson estimator (Falco J. Bargagli-Stoffi
101 et al., 2025), which employs both the marginal probability of an individual ik to be exposed
102 to the treatment level (w, g) $\pi_{ik}(w, g)$ and the joint probability of two individuals ik and
103 jk' to be simultaneously exposed to the treatment levels (w, g) and (w', g') , respectively,
104 $\pi_{ikjk'}(w, g; w', g')$.

105 NetworkCausalTree relies on a flexible in-sample splitting criterion function that can potentially
106 accommodate more than one causal estimand simultaneously: researchers rule the extent
107 of which the four estimands contribute to the objective function used to partition the
108 covariate space by assigning customized weights, included in the vector $\omega(w, g; w', g')$.
109 NetworkCausalTree procedure is divided into two steps: discovery and estimation. Each
110 cluster is randomly assigned to either the discovery or the estimation set (using a procedure
111 that is commonly referred to as *honest splitting*). In the discovery step, NetworkCausalTree
112 builds a tree according to the in-sample splitting criterion and stops when either the tree has
113 reached its maximum depth or any additional split would generate leaves that are not sufficiently
114 representative of the four exposure conditions. In the estimation step, NetworkCausalTree

estimates the CACEs and their standard errors in each leaf. The steps of the NetworkCausalTree algorithm are illustrated in the Algorithm 1 table.

Image of the Network Causal Tree Algorithm.

Figure 1: Image of the Network Causal Tree Algorithm.

Usage

NetworkCausalTree is available on [GitHub](#) and can be installed and loaded into the R session using:

```
library(devtools)
install_github("fbargaglistoffi/NetworkCausalTree", ref="master")
```

`data_generator_direct()` and `data_generator_direct_indirect()` are flexible synthetic dataset generators, which can be used for simulations before applying NetworkCausalTree to real-world data sets. Both accept key arguments `N` (sample size), `M` (number of binary covariates), `k` (number of clusters), `p` (vector of individual treatment assignment probabilities), `het` (whether treatment effects are heterogeneous), `h` (effect magnitude), and `method_networks` (network generation model). Both return a CNI environment, where cluster-specific networks are generated either using the Erdos-Renyi model (Erdős & Rényi, 1959), the Barabasi-Albert (Barabási & Albert, 1999), or the exponential random graph model (Lusher et al., 2013). `data_generator_direct()` generates outcomes with direct treatment effects only, while `data_generator_direct_indirect()` additionally includes spillover (indirect) effects in the outcome model. For `data_generator_direct_indirect()` with `het = TRUE`, both the direct effect $\tau_{(1,0;0,0)}$, and the spillover effect $\tau_{(0,1;0,0)}$ equal h if $x_1 = 0$ and $-h$ if $x_1 = 1$.

The `data_generator_direct()` and `data_generator_direct_indirect()` functions return as output a list of synthetic data including the covariates matrix (X), the outcome vector (Y), the individual intervention vector (W), the adjacency matrix (A), the neighborhood intervention vector (G), the group membership vector (K) and the vector describing the probability to be assigned to the active individual intervention vector (p).

```
dataset_direct <- data_generator_direct(N = 4000,
                                       M = 4,
                                       k = 80,
                                       p = rep(0.2,4000),
                                       het = TRUE,
                                       h = 2,
                                       method_networks = "er",
                                       param_er = 0.1)

dataset_direct_indirect <- data_generator_direct_indirect(N = 4000,
                                                         M = 4,
                                                         k = 80,
                                                         p = rep(0.2,4000),
                                                         het = TRUE,
                                                         h = 2,
                                                         method_networks = "er",
                                                         param_er = 0.1)
```

We propose here two examples of how to run the Network Causal Tree algorithm by the NetworkCausalTree package. To run the NetworkCausalTree function, the user has to input the following arguments: the data $(W,Y,X,M) = (W_{ik}, Y_{ik}, X_{ik}, M_{ik})$, where $ik \in \mathcal{V}$; the vector collecting individual treatment assignment probabilities p , the global adjacency matrix A that includes the cluster specific matrices A_k . Moreover, users must specify the weights vector

142 $\omega(w, g; w', g')$, ruling the extent of which each causal estimands contribute to the objective
 143 function (effect_weights); the ratio_disc parameter, representing the ratio of clusters to be
 144 included in the discovery set only; two parameters ruling the stopping criteria of the tree (depth
 145 measures the maximum depth, while minsize specifies the minimum number of observations
 146 for each level of the joint treatment (w, g) to be required in the leafs; the method to compute
 147 the objective function, that is, method = "singular" for NCT targeted to one single effect,
 148 method = "composite" for NCT targeted to multiple effects, method = "penalized" for a
 149 criterion function computed while considering a single effect only and including a penalization
 150 term related to the variance within the leafs; the desired output of the function (if output =
 151 "detection" only point estimates are computed, if output = "estimation" both estimated
 152 effects and their standard errors are computed).

153 **Example 1.** Running Network Causal Tree while relying on a singular in-sample splitting
 154 criterion function, that assesses the heterogeneity of the main treatment effect ($\tau_{(1,0;0,0)}$) only

```
result <- NetworkCausalTree(X = dataset_direct[["X"]],
                           Y = dataset_direct[["Y"]],
                           W = dataset_direct[["W"]],
                           A = dataset_direct[["A"]],
                           K = dataset_direct[["K"]],
                           p = dataset_direct[["p"]],
                           effect_weights = c(1,0,0,0),
                           ratio_disc = 0.5,
                           depth = 2,
                           minsize = 5,
                           method = "singular",
                           output = "estimation")
```

155 **Example 2.** Running Network Causal Tree while relying on a composite in-sample splitting
 156 criterion function, which accounts for all the four estimands simultaneously

```
result <- NetworkCausalTree(X = dataset_direct_indirect[["X"]],
                           Y = dataset_direct_indirect[["Y"]],
                           W = dataset_direct_indirect[["W"]],
                           A = dataset_direct_indirect[["A"]],
                           K = dataset_direct_indirect[["K"]],
                           p = dataset_direct_indirect[["p"]],
                           effect_weights = c(0.25,0.25,0.25,0.25),
                           ratio_disc = 0.5,
                           depth = 2,
                           minsize = 5,
                           method = "composite",
                           output = "estimation")
```

157 The results are included in a data.frame object which provides information about i) the nodes
 158 identified by the tree (FILTER), and their role in the tree - parent or leaf - (TERMINAL), ii) the
 159 number of units included in all nodes, both in the discovery set (NOBS_TR) and in the estimation
 160 set (NOBS_EST), iii) the estimated CACE's (EFF($w, g; w', g'$))_EST), with their corresponding
 161 standard errors (SE($w, g; w', g'$))_EST), in all nodes. The function print() displays these results
 162 and the plot_NCT() function visualizes the tree with the estimated effects in each leaf. In each
 163 node, we display, from top to bottom, the estimates of the main treatment effect, $\tau(1, 0; 0, 0)$
 164 and of the main spillover effect, $\tau(0, 1; 0, 0)$, and the size of the selected subpopulations. The
 165 colors indicate the sign and magnitude of either the main treatment effect, $\tau(1, 0; 0, 0)$, or
 166 the main spillover effect, $\tau(0, 1; 0, 0)$. Blue nodes correspond to positive estimated treatment
 167 effects, while green nodes represent positive estimated spillover effects. The intensity of the
 168 color indicates the strength of the effect: darker shades correspond to stronger effects, and
 169 negative effects are shown in white.

170 For instance, Figure 2 shows the tree obtained from **Example 1**. The code used to produce
171 the plot is as follows:

```
title <- expression(paste("CAUSAL TREE TARGETED TO ", tau, "(1,0;0,0)"), sep="")
cov_names <- colnames(dataset_direct[["X"]])

plot_NCT(NCT = result,
         cov_names = cov_names,
         title = title)
```

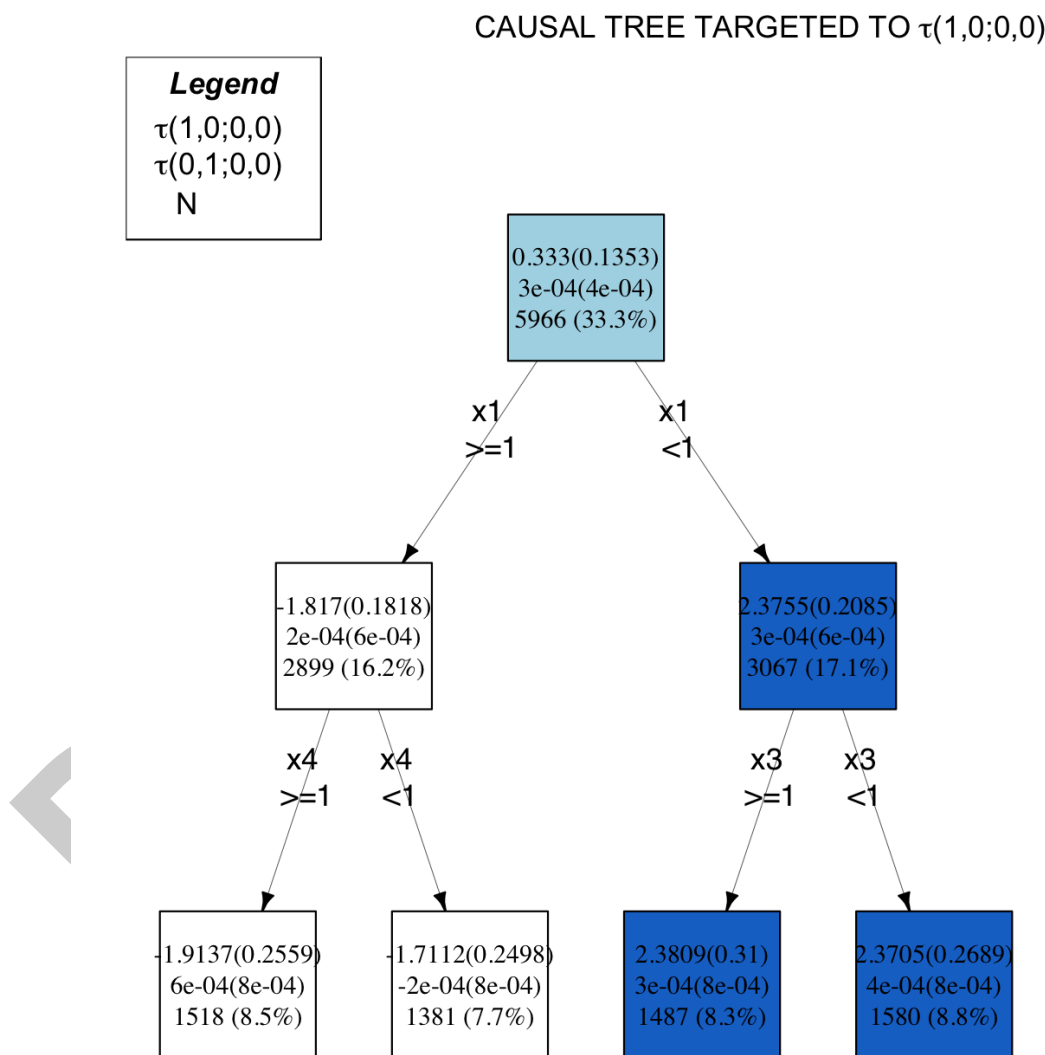


Figure 2: Network Causal Tree targeted to the main treatment effect

172 Figure 3 shows the tree obtained from **Example 2**.

```
title <- expression(paste("CAUSAL TREE TARGETED TO ", tau, "(0.25,0.25;0.25,0.25)"), sep="")
cov_names <- colnames(dataset_direct_indirect[["X"]])
```

```
plot_NCT(NCT = result,  
         cov_names = cov_names,  
         title = title)
```

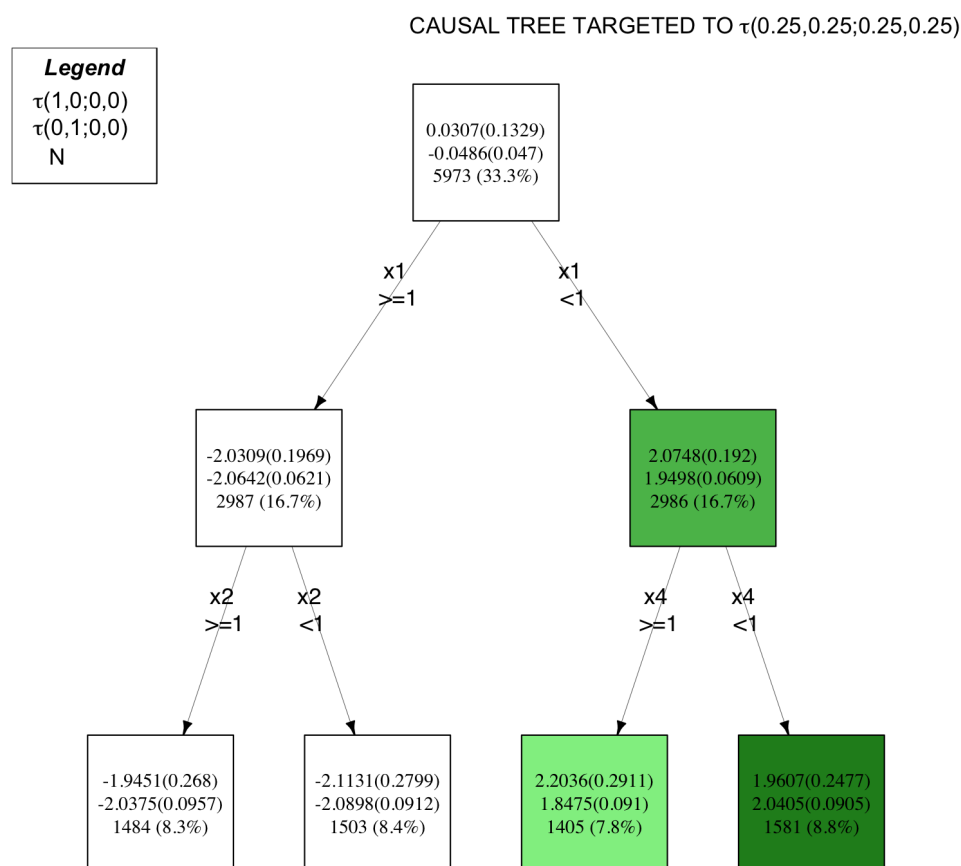


Figure 3: Network Causal Tree targeted to all effects

In these examples, the most important heterogeneity driver is x_1 . The estimated main treatment effect is strongly positive and close to 2 if $x_1 = 0$, while it is strongly negative and close to -2 for $x_1 = 1$ (note that this result is coherent with respect to the previous definition of the parameters related to the `data_generator_direct_indirect()` function, $h = 2$ and $het = TRUE$). The spillover effect is similarly heterogeneous, with $\tau_{(0,1;0,0)} = 2$ for $x_1 = 0$ and $\tau_{(0,1;0,0)} = -2$ for $x_1 = 1$.

Online documentation for the package can be found at f bargaglistoffi/NetworkCausalTree.

Reproducibility and dependencies

All examples in this paper can be reproduced using the scripts provided in the `tests/examples/` directory of the package's GitHub repository. The package relies solely on standard R libraries, which are automatically installed and loaded upon installation of `NetworkCausalTree`. Numerical stability and reproducibility have been validated through extensive simulation studies and example analyses across multiple systems. **The package passes all standard R CMD check tests without errors or warnings, confirming full compliance with CRAN submission standards.**

Implementation and future work

This package is implemented in R and follows standard R development conventions, ensuring compatibility across systems. The GitHub repository provides example scripts to facilitate reproducibility. Community engagement is encouraged through documented contribution guidelines ([CONTRIBUTING.md](#)) and a code of conduct ([CODE_OF_CONDUCT.md](#)). Users can report issues or request support through the GitHub Issues page.

Future extensions will include support for continuous exposures (e.g., multiple dosage levels of a vaccine), integration with random forest estimators for ensemble-based causal discovery (similar to how `grf` builds on `causalTree`), and diagnostic tools for sensitivity analysis under partial interference (e.g., assessing whether the CNI assumption is reasonable given the data).

AI usage disclosure

AI-assisted tools were used in two capacities: (i) GitHub Copilot was used to assist with code review; (ii) a large language model was used for copy-editing and proofreading of the paper text. Technical content, software design decisions, algorithmic choices, and methodological contributions were made by the human authors, who reviewed and validated AI-assisted outputs.

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